

# Automated Medical Podcast Transcription and Topic Segmentation

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## 1. Introduction

In recent years, medical podcasts have become an important source of information for medical students, healthcare professionals, and researchers. These podcasts include discussions on diseases, treatments, clinical experiences, and public health awareness. However, most medical podcasts are long in duration, making it difficult for listeners to quickly identify relevant topics.

Listening to an entire podcast to find a specific discussion is time-consuming and inefficient. To overcome this problem, there is a need for an intelligent system that can automatically transcribe medical podcasts, divide them into meaningful topic segments, and provide summaries, keywords, and sentiment insights.

The **Automated Medical Podcast Transcription and Topic Segmentation** project addresses this challenge by using Artificial Intelligence (AI) and Natural Language Processing (NLP) techniques to convert medical audio into structured, searchable, and easy-to-understand content.

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## 2. Problem Statement

Medical podcasts contain valuable information, but their long duration makes it difficult for users to locate specific medical topics quickly. There is no efficient mechanism to navigate podcasts based on diseases, symptoms, or treatments discussed within them.

Therefore, the problem is to design a system that can:

- Automatically convert medical podcast audio into text
  - Identify topic boundaries within the transcript
  - Generate summaries, keywords, and sentiment analysis
  - Provide a user-friendly interface for navigation and analysis
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## 3. Objectives of the Project

The main objectives of this project are:

1. To convert medical podcast audio into accurate text using speech-to-text models
2. To segment the transcript into meaningful medical topics

3. To extract important medical keywords from each segment
4. To analyze sentiment and quality metrics of the transcription
5. To provide an interactive frontend for viewing transcripts, segments, and analytics

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#### 4. Scope of the Project

The scope of this project includes:

- Processing medical podcast audio files
- Automatic transcription using ASR models
- Topic segmentation using NLP techniques
- Keyword extraction and sentiment analysis
- Quality evaluation using WER, CER, and semantic similarity
- Visualization through a web-based frontend

The project does not include live streaming audio or real-time transcription.

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#### 5. System Architecture

The system follows a modular, pipeline-based architecture:

Audio Input

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Audio Preprocessing

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Medical Speech-to-Text (ASR)

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Transcript Cleaning

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Embedding Model

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Topic Segmentation

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Summaries, Keywords & Sentiment

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Quality Evaluation

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Indexing

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Frontend User Interface

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## **6. Module Description**

### **6.1 Audio Preprocessing Module**

This module prepares raw medical audio for transcription. It includes audio normalization and noise handling to improve transcription accuracy.

### **6.2 Transcription Module**

The transcription module converts audio into text using advanced ASR models such as Whisper or Faster-Whisper. It also generates timestamps for each segment.

### **6.3 Topic Segmentation Module**

This module identifies topic changes within the transcript using NLP techniques such as text similarity and embedding models. Each segment represents a meaningful medical discussion.

### **6.4 Keyword Extraction Module**

Important medical terms are extracted from each segment using keyword extraction algorithms. These keywords help users quickly understand the topic.

### **6.5 Sentiment and Quality Evaluation Module**

This module performs:

- Segment-level sentiment analysis
- Normalization of sentiment scores for medical context
- Quality evaluation using:
  - Word Error Rate (WER)
  - Character Error Rate (CER)
  - Semantic similarity

### **6.6 Frontend Visualization Module**

The frontend displays:

- Full transcript
- Segmented summaries

- Keyword clouds
- Sentiment graphs
- Quality dashboards

Users can navigate topics using timestamps and search for specific medical terms.

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## **7. Technology Stack**

### **Backend Technologies**

- Python 3.9+
- Flask Framework
- Whisper / Faster-Whisper
- Librosa, PyDub, FFmpeg

### **NLP and Machine Learning**

- NLTK
- SpaCy
- HuggingFace Transformers
- Sentence Transformers
- KeyBERT / YAKE / RAKE

### **Frontend Technologies**

- React.js
- HTML, CSS, JavaScript
- REST APIs

### **Visualization Tools**

- Chart.js
- Plotly
- Matplotlib

### **Storage**

- JSON and CSV files
- SQLite (optional)

- FAISS (optional)
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## **8. Project Structure**

The project is organized into backend and frontend modules:

- Backend (src/) contains preprocessing, transcription, segmentation, sentiment, and evaluation logic
  - Frontend (ui\_app/) contains React components for dashboards and visualization
  - Inference folder stores generated transcripts and outputs
  - Documentation and testing folders support development and evaluation
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## **9. Milestone-wise Implementation**

### **Milestone 1: Audio Preprocessing & Transcription**

- Audio normalization
- ASR-based transcription

### **Milestone 2: Topic Segmentation & Keyword Extraction**

- Topic boundary detection
- Medical keyword extraction

### **Milestone 3: Sentiment & Quality Evaluation**

- Segment-level sentiment analysis
- WER, CER, and semantic similarity computation

### **Milestone 4: Frontend Integration & Visualization**

- React-based UI
- Topic navigation and playback
- Keyword, sentiment, and quality dashboards

### **Milestone 5: Documentation & Final Delivery**

- Technical documentation
  - Final project demonstration
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## 10. Data and Privacy Considerations

- Raw audio files are not stored in the repository
  - API keys are handled using environment variables
  - Only source code and configuration files are version-controlled
  - User data is not permanently stored
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## 11. Future Enhancements

- Medical Named Entity Recognition (NER)
  - Speaker diarization
  - Semantic search across multiple podcasts
  - Multi-language transcription support
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## 12. Intended Users

- Medical students
  - Healthcare professionals
  - Medical researchers
  - Medical educators
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## 13. Conclusion

The **Automated Medical Podcast Transcription and Topic Segmentation** project provides an efficient solution for navigating long medical podcasts. By combining AI, NLP, and frontend visualization, the system makes medical audio content more accessible, searchable, and informative. This project demonstrates the practical application of AI technologies in the healthcare education domain.

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## 14. License

This project is licensed under the **MIT License**.